Distortion Robust Relative Radiometric Normalization of Multitemporal and Multisensor Remote Sensing Images Using Image Features

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***Abstract*— In this article, we propose a novel framework to radiometrically correct unregistered multisensor image pairs based on the extracted feature points with the KAZE detector and the conditional probability (CP) process in the linear model fitting. In this method, the scale, rotation, and illumination invariant radiometric control set samples (SRII-RCSS) are first extracted by the blockwise KAZE strategy. They are then distributed uniformly over both textured and texture-less land use/land cover (LULC) using grid interpolation and a set of nearest-neighbors. Subsequently, SRII-RCSS are scored by a similarity measure, and the histogram of the scores is then used to refine SRII-RCSS. The normalized subject image is produced by adjusting the subject image to the reference image using the CP-based linear regression (CPLR) based on the optimal SRII-RCSS. The registered normalized image is finally generated by registration of the normalized subject image to the reference image through a two-pass registration method, namely affine-B-spline and, then, it is enhanced by updating the normalization coefficient of CPLR based on the SRII-RCSS. In this study, eight multitemporal data sets acquired by inter/intra satellite sensors were used in tests to comprehensively assess the efficiency of the proposed method. Experimental results show that the proposed method outperforms the existing state-of-the-art relative radiometric normalization (RRN) methods both qualitatively and quantitatively, indicating its capability for RRN of unregistered multisensor image pairs.**

***Index Terms*— Change detection, feature detection, feature matching, image matching, relative radiometric normalization, rotation invariant, scale invariant.**

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I. Introduction

THERE is no doubt that with the advancement of technology, access to remote sensing data acquired from different sensors has been increased such that they are nowadays an imperative source of information in earth science disciplines [1], [2]. The spectral information of satellite images is negatively affected by various types of radiometric distortions due to a variety of factors, such as sensitivity to atmospheric conditions (e.g., scattering and absorption effects), temporal and spatial variations in the sensor-target-illumination geometry, and sensor calibration errors [3]–[5]. Consequently, radiometric correction is the main step for various remote sensing applications, such as time series analysis [6], unsupervised change detection [7], image mosaicking [5], [8], [9], and environmental monitoring [10].

Generally, there are two broad categories of radiometric correction approaches [11]: absolute radiometric normalization (ARN) and relative radiometric normalization (RRN). ARN approaches typically convert digital numbers (DNs) of pixels to surface reflectance values through modeling atmosphere or definite transformation equations [12]. These methods require costly ground-based measurements at the time of imagery and sensor calibration and, therefore, are impractical when dealing with archived remote sensing data [4], [12], [13]. On the other hand, RRN techniques are often preferred due to the use of affordable image processing operations to rectify radiometric distortions in temporal images without having sensor parameters and *in situ* atmospheric data at the time of imagery acquisition [13], [14]. These techniques rectify the radiometric characteristics of a multiband subject image to a multiband reference image, which is usually selected by statistical and visual analyses [4], [15].

Depending on how the pixel pairs between the reference and subject images are selected, RRN techniques can be divided into two categories [15]: dense RRN (DRNN) and sparse RRN (SRRN). The DRRN methods, such as haze correction

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(HC) [16], and histogram matching (HM) [17], employ the global reference-subject image pair statistics without considering the land use/land cover (LULC) changes and anomalies for radiometric rectification of the subject image [18], [19]. Although DRRN techniques are often computationally effective, they are only suitable for an RRN of reference-subject image pair with the small LULC changes [15]. Therefore, the success rate of these methods is strongly dependent on the amount of LULC change between the subject and reference images. However, SRRN techniques have the potential to precisely estimate the relationship between the reference and subject images with/without LULC changes using the time-invariant pixels as the pseudo-invariant features (PIFs) or radiometric control set samples (RCSS) [13], [15], [18]. The accuracy of SRRN is highly dependent on the quality, quantity, and spatial distribution of the RCSS.

A variety of strategies have been so far developed to select optimal RCSS. For example, Canty *et al.* [20] employed multivariate alteration detection (MAD) transformation [21] to automatically extract the RCSS from a reference-subject image pair. The extracted RCSS was then utilized to create a relationship between reference and subject images through a linear regression model. The efficacy of this method is negatively affected in dealing with data sets which include many LULC change (e.g., large seasonal changes in vegetation) [22], [23]. Following this method, Canty and Nielsen [22] identified the iteratively re-weighted modification of MAD transformation (IRMAD) as one of the widely used techniques to cope with temporal LULC changes in RRN. For example, a parallel SRNN method based on the IRMAD transformation and orthogonal regression was presented by Chen *et al.* [24] to handle massive remote sensing images. The results showed that the presented method achieved good color consistency between multiple images by considering the effect of the normalization path for calculating normalization coefficients [5]. Zhang *et al.* [25] proposed an automatic SRRN method to reduce radiometric distortions between image pairs based on the iterative slow feature analysis (ISFA). In this method, reformulated SFA was first employed to select the RCSS from image pair in an iterative weighted process such that high weights are allocated to unchanged samples in each iteration. The relationship between image pairs was then directly determined through a linear regression based on all the pixels and their weights. If the image pairs had dominated changed regions, initial seeds were utilized as an alternative for the initial weights in the iteration process to improve the efficacy of ISFA, which is named S-ISFA. The results indicated that the ISFA and S-ISFA had better performance than the IRMAD in adjusting the radiometric differences. Kim *et al.* [26] improved an SRRN approach for multitemporal hyperspectral data by automatically selecting RCSS using the cumulative average of the spectral angle mapping (SAM) and its differential values. The subject image was finally adjusted to the reference image through an iterative linear regression based on the selected RCSS. The reliability of this method was dependent on the quantity of the selected RCSS. Sadeghi *et al.* [18] proposed an SRRN method based on the unchanged pixels resulted from simple unsupervised change detection and a nonlinear mapping function, estimated by an artificial neural network (ANN). Although this method had high flexibility in modeling the relationship between image pairs, it was not computationally efficient and required to adjust the number of neurons in the hidden layers of the ANN for optimal performance [15]. Lin *et al.* [27] also proposed an SRRN method by selecting RCSS using weighted principal component analysis (PCA) of the scattergram of the reference and subject images. The main disadvantage of this method was dealing with the image pairs which contained clouds. Moreover, Zhou *et al.* [19] employed the normalized difference water index (NDWI) to extract the dark and bright sets of RCSS. They also controlled the fraction of RCSS by statistical rules in an iterative process. The optimal RCSS was then used to model the linear relationship between reference and subject images through linear regression. Although the method had promising results in RRN of multiple images acquired by inter/intra sensors, its performance was negatively affected by seasonal and interannual phenological variations [19]. Seo *et al.* [28] also proposed a robust RRN method to reduce the radiometric and phenological differences in multitemporal images based on the Random Forest (RF) regression and RCSS selection from scattergrams used in a no-change (NC) regression [29]. The proposed method was later improved by Seo and Eo [14] using a nonlinear regression model-multilayer perceptron (MLP) instead of RF regression, as well as considering the spectral indices to compensate for the phenological properties in selecting RCSS. The two aforementioned methods were limited when reference and subject images did not contain both water and land covers because the proposed methods only employed the centers of these LULC in the scattergrams to determine the RCSS [30]. Furthermore, Syariz *et al.* [23] proposed a spectral-consistent SRRN method to preserve the consistency of the spectral signature of LULC classes during the RRN procedure. For this, a constrained orthogonal regression was developed to approximate the relationship between the RCSS which are selected by the IRMAD method from the image pair. To better handle the possible spectral variations during RRN, the middle of the radiometric levels of the image pair was also considered as a reference instead of taking one of them. The experimental results showed that the presented constrained regression had superior performance to ordinary and orthogonal regressions. However, this method was only improved on the Landsat 8 image pairs, so it is possible that its constraints are not sufficient for the spectral consistency issues in cross-sensor RRN [23]. Moghimi *et al.* [15] developed a multistep strategy to automatically select robust and well spatially distributed RCSS using fast local Laplacian filter (FLLF)-based difference image, the improved fast level set method (IFLSM) [31], and a patch-based outlier detection method. Although this method provided better accuracy compared to several well-known RRN methods, it was computationally expensive. More recently, Zhang *et al.* [32] identified a block adjustment-based radiometric normalization (BARN) method to adjust multiple remote sensing images by taking into account the global and local radiometric distortions. In this method, a block adjustment-based RNN was first employed to remove the global radiometric distortions and

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reduce the error accumulation and adaptive allocation of cumulative errors, resulting from transfer paths. To eliminate the local residual radiometric distortions, two local approaches based on block adjustment and edge optimization were then adopted. The results demonstrated that the BARN method outperformed the momentum match (MM) and the HM algorithm (i.e. presented in ENVI software) in terms of accuracy and visual inspection.

Although the aforementioned methods provide promising RRN results for multitemporal satellite images acquired by a specific remote sensing system, they are not applicable to multisensor satellite images. Most of the current RRN methods select RCSS based on the similarity between pixel intensities of the reference-subject image pair, which is not satisfactory for RRN of image pair acquired by different sensors with differing acquisition and illumination geometries. Moreover, the differences between the spectral band position in multisensor image pair are rarely considered in the RRN procedure, while this potentially leads to a significant difference in spectral signatures of LULC classes. Furthermore, the SRRN methods cannot handle the radiometric differences between image pairs with different spatial resolutions, unless they are resampled to the same resolution. The impact of possible outliers in the linear model fitting process is also less regarded while this can notably affect the RRN results [33]. Finally, the co-registration of the subject and reference images is a mandatory prerequisite for SRNN methods [15] and, thus, the application of these methods is limited when RRN of the unregistered multisensor image pair is required. In this regard, HM, as a cost-effective DRRN method, is the only technique that can be employed for radiometric correction of the unregistered image pair. In this method, the reference and subject images do not need to have the same spatial resolution and size for RRN. However, the reliability of this method is affected by the LULC change and misalignment between reference and subject images [18], [19].

In this study, a novel SRRN method was proposed to address the RRN limitations in unregistered multitemporal/multisensor satellite images with the capability of accurately working with images with different spatial resolutions. The proposed method utilized the KAZE detector [34] to automatically extract the scale, rotation, and illumination invariant RCSS (SRII-RCSS) from a reference-subject image pair. The KAZE employed non-linear diffusion filtering for reconstructing scale-space and the Modified Speeded Up Robust Features (M-SURF) descriptor [35] for key points extraction. This resulted in better preservation of image detail (e.g., boundary) compared to the scale-invariant feature transform (SIFT) [36] and original SURF [37] detectors. Moreover, the conditional probability (CP) was utilized to accurately estimate a linear relationship between multisensor image pairs. In summary, compared with the aforementioned state-of-the-art RRN methods, our proposed method had the following contributions.

1) A feature-based method (i.e., KAZE detector) was used for the first time for automatic SRII-RCSS extraction to address illumination differences and misregistration errors in RRN of the multitemporal/multisensor image pairs. This strategy also handled the large differences in spatial resolutions and rotations between multisensor reference-subject image pairs during the RRN procedure.
2) Although using match points seems to be sufficient in the registration process, their lack in texture-less LULC regions (e.g., water bodies) leads to errors in the RRN results. To tackle this problem, a grid based interpolation and a set of nearest neighbors' points were employed in this study to increase SRII-RCSS in both textured and texture-less areas. A score index based on correlation and Chebyshev distance was also proposed to detect the possible inappropriate SRII-RCSS.
3) A robust linear regression was proposed based on the CP process to accurately estimate normalization coefficients. This approach eliminated the possible outliers from the fitting procedure.

The remainder of this article is structured, as follows. Section II describes the proposed SRRN method, details of the eight used data sets, and evaluation metrics. Section III provides the experimental results for RRN of unregistered and registered image pairs, and finally, concluding remarks are presented in Section IV.

II. Materials and Methods

A. Proposed SRRN Method

Let us consider the reference and subject images with different/similar sizes were acquired by intra/inter sensors over the same geographical region at different times. The main objective of this study was to generate an optimal normalized subject image from the reference-subject image pair, acquired by inter/intra sensors, based on reliable and well-distributed SRII-RCSS. As illustrated in Fig. 1, the proposed RRN method was applied to two scenarios for RRN of unregistered and registered image pairs, and composed of five main steps, including: 1) coarsely estimating the transformation parameters between the subsampled subject-reference image pair from sparse matches between them, extracted using the KAZE; 2) extracting the well spatially distributed matches as SRII-RCSS based on a block-wise KAZE strategy with the help of the transformation parameters estimated in the first step; 3) evenly distributing the SRII-RCSS in both textured and texture-less LULC classes using the grid-based interpolation and a set of nearest neighbors, and refined them from possible changed/invariance samples based on a similarity measure; 4) generating a normalized subject image through the proposed CP-based Linear Regression (CPLR) method based on the refined SRII-RCSS; and 5) improving the normalized subject image by registering that based on a combination of rigid and non-rigid registration algorithms (i.e., affine-B-spline) and updating its normalization coefficients using the CPLR method. The details of the steps are explained in the following subsections.

1) Step 1: Coarse Spatially Alignment: In the first step, the relative transformation parameters between the subject and reference images were estimated using a few matched feature points for further analysis. For this purpose, the reference and subject images were first down-sampled to reduce the

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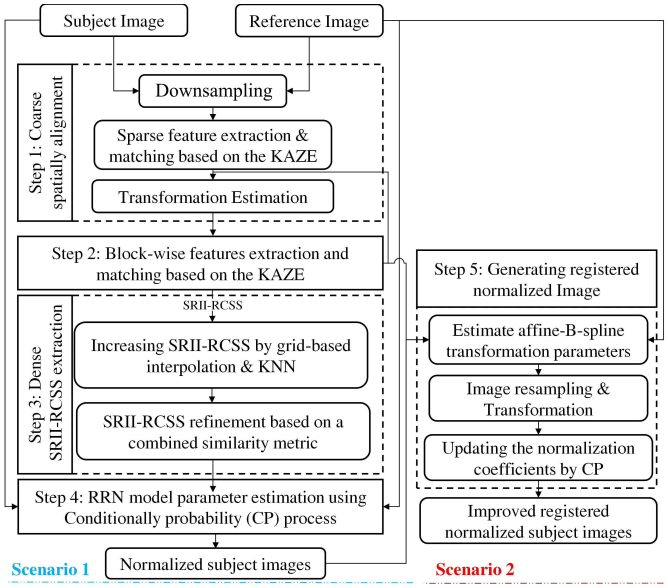

Fig. 1. Workflow of the proposed RRN method.

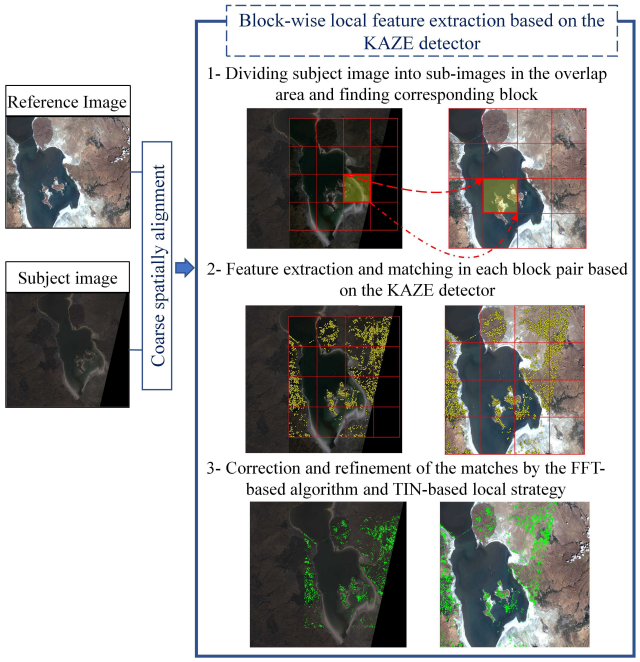

Fig. 2. Main steps of the match points extraction using blockwise strategy.

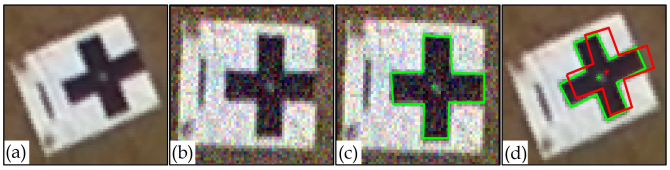

Fig. 3. Example of performance of the FFT-based algorithm to correct the position of match points. A squared target in the (a) subject and (b) reference images. The position of the match points, manually selected on the squared target in (c) subject image and (d) reference image before (in red) and after (in green) correction by the FFT-based technique.

computation costs, and the KAZE detector was then applied to extract sparse feature points from down-sampled images. The matched features were subsequently identified based on the nearest neighbor distance ratio method as proposed in [36]. Finally, The RANdom SAmple Consensus (RANSAC) [38] algorithm was applied to remove the false matches, and the affine transformation parameters were relatively estimated based on the refined match points. It is worth mentioning that the down-sampling led to the loss of image details and reduced the quality and quantity of the match points. However, it has been proved that this approach could be efficiently employed to generate sparse feature points that were relatively corresponding to each other, and could be sufficient to coarsely estimate the transformation parameters [39], [40]. Moreover, a weighted sum of the three bands [e.g., red, green, and blue (RGB), and green, red, and near-infrared (GRNIR)] of image pair was used for performing the feature extraction faster.

2) Step 2: Block-Wise Feature Extraction and Matching: The number and spatial distribution of matched features are important for the accurate image registration [41] and can affect the RRN results. Therefore, in this step, dense and well spatially distributed local features were extracted from the reference-subject image pair using a block-wise strategy (see Fig. 2). The overlap area in the reference and subject images was first computed based on the geometrical constraints provided by the reference image coordinate and the approximated transformation in the previous step. The overlapped area in the subject image was then subdivided into multiple blocks, and the corresponding blocks in the reference image were identified by transforming the four corners of each block into the reference image based on the approximated transformation. The KAZE detector was then employed to extract more robust-illumination and well spatially distributed features from each aligned block.

After extracting the match feature points from each block pair, they should be refined from outliers. Although the RANSAC algorithm can be used as a post-process KAZE matching to keep the most precise match points, this process removes a large number of matches that can be potentially corrected. Therefore, in this study, the RANSAC method with a large threshold value (5 pixels were used in this study) was first applied to remove only the number of mismatches with the large position error. A window with specific dimensions (9×9 was used in this study) was considered for each remaining corresponding points in the image pair and, then, the FFT-based cross correlation algorithm [42] by an integer factor of up-sampling was applied to accurately estimate the transform parameters between them. Finally, the position of the local features in the subject image is corrected by using the corresponding estimated transform parameters. For example, the performance of the FFT-Based algorithm to correct the position of the corresponding points was tested on a squared target in reference-subject image pair (see Fig. 3).

As demonstrated in Fig. 3, although the subject image was noisy, the position of its corresponding point in the reference image was accurately corrected using the FFT-based technique, indicating its high potential in modeling residual positional errors (e.g., differences in rotation, transition, and scale)

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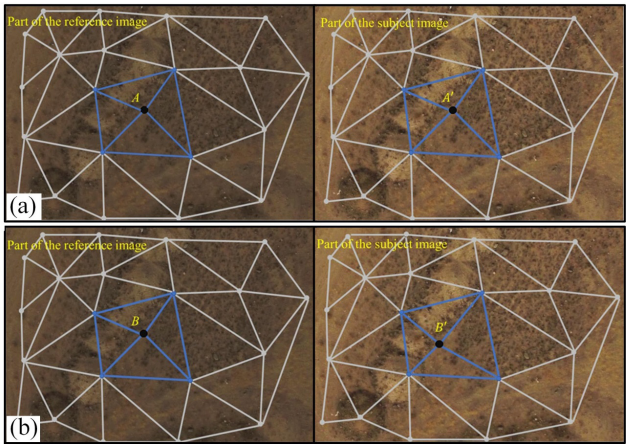

Fig. 4. Procedure of wrong match detection. (a) Correct match point (A, A') and (b) false match point (B, B').

between match points. Although the position of a large number of the matches was corrected using the FFT-based method, it was still possible that there were several false matches due to the existence of various complex distortions in the images captured by different sensors. Since these distortions cannot be completely handled by a global approach, a local strategy based on the triangulated irregular network (TIN) structure and affine transformation as presented in [43] was employed for outlier refinement. Using this approach, a TIN was first constructed in the subject and reference image based on the coordinates of match points. The local affine parameters were then determined for each of the match points using several nearest neighborhood points in an iterative procedure. Finally, the residual error was calculated for each of the matches using the corresponding local affine parameters. If the residual error was larger than a certain threshold (1 pixel was used in this study), the corresponding feature points were considered as outliers and were removed from further analysis. The analysis of the false match detection by the above procedure is illustrated in Fig. 4.

As shown in Fig. 4, by examining the triangles generated by neighborhood points, the positional difference and residual errors at the correct and wrong match points were, respectively, very small and big. Thus, the outliers could be accurately identified and removed. For more information about error elimination through the local estimation, it is recommended to refer to [43]. After eliminating false matches, the remaining matches were used as SRII-RCSS for further analyses of RRN and estimating the affine-B-spline transformation parameters in the registration procedure.

3) Step 3: Dense SRII-RCSS Extraction: Although the quantity and spatial distribution of the detected match points generated from Step 2 were enough for the registration procedure, a few of them were in the homogeneous or texture-less LULC areas, such as water bodies and the deserts, which could negatively affect the RRN results. This is mainly due to the nature of these methods, which use distinct objects (e.g., corners or blobs) for feature detection. To resolve this problem, the numbers of SRII-RCSS in the texture-less areas were increased using grid interpolation. To this end, a regular grid of

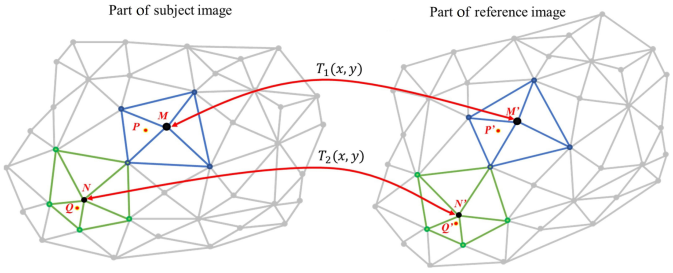

Fig. 5. Process of identifying the position of each grid point (P and Q) in the reference image based on the estimated local affine transformations of the nearest SRII-RCSS (M, N).

points was first created based on the position of the SRII-RCSS across the whole overlap area in the subject image. The local affine transformation (which was calculated at step 2) of the closest SRII-RCSS (corresponding points) to each grid point was employed to estimate the exact position of the points in the reference image. For example, as illustrated in Fig. 5, if (M, M') and (N, N') be SRII-RCSS pair (corresponding points) in the subject and reference images, the position of grid points P and Q in the reference image can be accurately calculated through the estimated local affine transformations of the nearest SRII-RCSS, which are, respectively, $T_1(x, y)$ and $T_2(x, y)$.

Given that the characteristics of each match point in an image are generally similar to the characteristics of adjacent points, a set of nearest points surrounding SRII-RCSS (matched and grid points) in the subject image was first considered by the K-nearest neighbors (KNN) algorithm. Their positions were then calculated in the reference image with the same strategy as for the grid points. Therefore, there was a cluster of SRII-RCSS in each position, which increased the reliability of the RRN procedure. Although a high number of radiometric control points can potentially indicate better overall performance, this can result in potential errors in RRN for the LULC change analysis applications [19]. Therefore, the spectral reflectance constancy of the SRII-RCSS should be controlled. To this end, a novel score index was proposed in this study to grade the SRII-RCSS in terms of their invariance properties which is calculated as follows:

$$U^j = \sqrt{\mathrm{Corr}(s^j,\ r^j).(1 - D_{\mathrm{Ch}}(s^j,\ r^j))}, \quad j = 1, 2, \ldots, n \tag{1}$$

in which (s^j, r^j) are SRII-RCSS pair from the subject and reference images, respectively; n is the total number of pairs; and Corr (.) and $D_{\mathrm{Ch}}(.)$ denote the correlation and Chebyshev distance metrics, respectively, which can be obtained from the following equations:

$$\mathrm{Corr}(s^j,\ r^j) = \frac{\sum_{i=1}^{b}\left(s_i^j - \bar{s^j}\right)\left(r_i^j - \bar{r^j}\right)}{\sqrt{\sum_{i=1}^{b}\left(s_i^j - \bar{s^j}\right)^2}\sqrt{\sum_{i=1}^{b}\left(r_i^j - \bar{r^j}\right)^2}} \tag{2}$$

$$D_{\mathrm{Ch}}(s^j,\ r^j) = \max_i\left(\left|s_i^j - r_i^j\right|\right) \tag{3}$$

where $\overline{s^j}$ and $\overline{r^j}$ are, respectively, the average of s^j and r^j in all spectral bands; and $i = 1,\ 2, \ldots, b$, (b is the number of

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bands). The structure information (dependencies) and intensity comparison were performed by correlation and Chebyshev distance metrics. The higher degree of correlation and the smaller value of Chebyshev distance mean more similarity between the SRII-RCSS pair and vice versa. It is worth noting that each metric was first rescaled with the min-max normalization process in the range [0, 1]. Using this score index helps to measure the quality of the SRII-RCSS from a variety of perspectives and, thus, were more reliable for further analysis. After scoring the SRII-RCSS, the appropriate SRII-RCSS candidates were obtained by thresholding the histogram of scores. Since the SRII-RCSS in texture-less areas (e.g., water and desert) relatively have higher scores, they inevitably affect the RRN results. Therefore, choosing an appropriate threshold is a very important task in our method. In this way, a simple quartile-based threshold, t [(4)], was proposed to extract the potential unchanged SRII-RCSS.

$$t = \frac{\mathrm{Med}(U) + \mathrm{MH}(U)}{2} \tag{4}$$

where Med(.) is a median operator and refers to the second quartile and MH(.) is the midhinge operator indicating the average of the first and third quartiles. Finally, the SRII-RCSS with a score of more than t was selected for further process. It is worth noting that, using (4) instead of the first quartile reduces the risk of entering the changed samples in the RRN procedure. Moreover, the proposed threshold selects the reliable SRII-RCSS from both texture-less and textured regions, where the third quartile leads to the selection of the SRII-RCSS in texture-less areas in most cases.

4) Step 4: RRN Model Parameter Estimation Using CP Process: Based on the generated potential unchanged SRII-RCSS, the normalized subject image (S^N) was then created by rectifying the subject image to the reference image through a general mapping function which is calculated as follows:

$$S_i^N = \alpha_i S_i + \beta_i \tag{5}$$

where S_i^N and S_i, respectively, denote the DN values of the normalized subject image and original subject image in ith spectral band; and α_i and β_i are normalization coefficients for the ith spectral band. Since the response of the linear model is highly depended on values of these coefficients, any error in their estimating can cause a considerable error in the RRN results. Consequently, inspired by [44], we proposed a CP-based method to estimate the accurate normalization coefficients for the mapping function [(5)]. For this purpose, let us consider the r_p and s_p are the DN values of the SRII-RCSS in the reference and subject images, respectively. The CP process with a specific significant level was utilized to estimate normalization coefficients. CP was employed because it has a high capability to remove outliers from SRII-RCSS candidates in the liner model fitting. Assuming a bivariate normal (BVN) distribution for the DNs of the SRII-RCSS pixel pair, the conditional distribution of the r_p given the s_p, is expressed as follows for each of the spectral bands:

$$f\left(r_p \middle| s_p\right) = \frac{1}{\sigma_r \sqrt{2\pi(1-\varsigma^2)}} e^{-0.5\left[\frac{(r_p - m_r) - \varsigma\frac{\sigma_r}{\sigma_s}(s_p - m_s)}{\sigma_r\sqrt{1-\varsigma^2}}\right]^2} \tag{6}$$

in which m_r and m_s are, respectively, the means of r_p and s_p; σ_r and σ_s are the standard deviations of r_p and s_p, respectively; and ς is the correlation between r_p and s_p. The defined CP follows normal distribution with the mean and standard deviation of $m_r + \varsigma(\sigma_r/\sigma_s)(s_p - m_s)$ and $\sigma_r\sqrt{1-\varsigma^2}$, respectively. In this CP model, the upper and lower critical regions can be defined based on the two following parallel lines:

$$\left[m_r + \varsigma\frac{\sigma_r}{\sigma_s}\left(s_p - m_s\right)\right] - k_\alpha(\sigma_r\sqrt{1-\varsigma^2}) > r_p$$

or

$$\left[m_r + \varsigma\frac{\sigma_r}{\sigma_s}\left(s_p - m_s\right)\right] + k_\alpha(\sigma_r\sqrt{1-\varsigma^2}) < r_p \tag{7}$$

where selection of the k_α is depending on the selected level of significance, α (e.g., for $\alpha = 0.01$, $k_\alpha = 2.575$). In fact, the objective was selecting SRII-RCSS, which were between these two parallel lines in each band to estimate normalization coefficients. Therefore, a simple hypothesis testing was defined in each spectral band as

$$\begin{cases} H_0 : \text{SRII-RCSS} = \text{optimal SRII-RCSS} \\ H_1 : \text{SRII-RCSS} = \text{outlier} \end{cases} \tag{8}$$

in which H_0 indicates the SRII-RCSS candidates, which are between two parallel lines. These candidates were selected as optimal SRII-RCSS for estimating the normalization coefficient; and H_1 refers to the SRII-RCSS candidates which are outliers. Therefore, the central line between two aforementioned parallel lines was finally considered as the mapping function between the optimal SRII-RCSS in the reference and subject images through (5). The orientation of this line can be obtained from the slopes of the parallel lines and is calculated as follows:

$$\alpha = \varsigma\frac{\sigma_r}{\sigma_s}. \tag{9}$$

After estimating the slopes for each spectral band based on the CP analysis, the intercept of each spectral band was estimated using

$$\beta = m_r - \alpha \times m_s. \tag{10}$$

Therefore, the slope and intercept, as normalization coefficients in (5), were obtained using (9) and 10, respectively. Hereinafter, we refer such a regression method to CPLR (CP-based linear regression). Fig. 6 illustrates the proposed CPLR in the RRN procedure.

As shown in Fig. 6, the samples that are far from the mass of the SRII-RCSS were refined using the upper and lower edges (blue lines) of the critical region of the CP process, and a robust regression line (green line) was created by the remaining SRII-RCSS. The generated SRII-RCSS were randomly divided into training samples (70%) and test samples (30%) to develop and evaluate the proposed RRN method, respectively.

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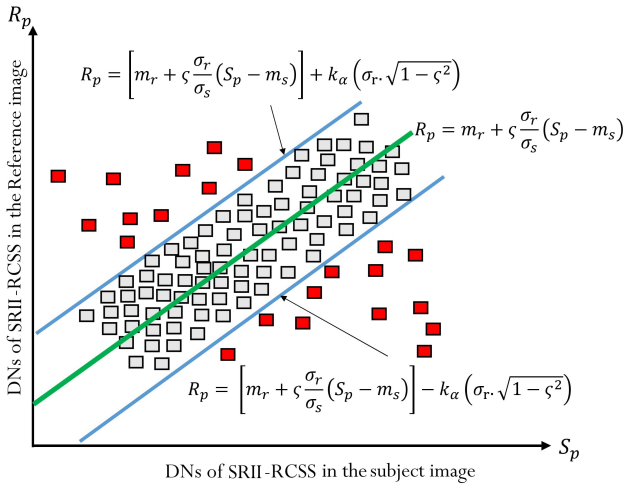

Fig. 6. Detected outliers (red symbols) from the SRII-RCSS by the proposed CPLR (blue lines refer to (7) and green line refer to regression line obtained from the (9) and (10).

5) Step 5: Generating the Registered Normalized Image: In the first scenario of the proposed method (see Fig. 1), the normalized subject image was generated without applying any co-registration operations. However, when the objective of using multisensor satellite images is time-series analyses or identifying changes, the image pair must be registered to each other which was addressed in the second scenario of the proposed method (see Fig. 1). In this regard, two methods were followed to generate a registered normalized image. In the first approach, the registered normalized image was produced by only registration of the generated normalized image at step 4 to the reference image. Using this approach, some of the radiometric differences could be occurred after the registration procedure due to the use of interpolation methods (e.g., bilinear) in the image warping operation. To resolve this issue, the registered normalized image was re-generated in the second method by adjusting it to the reference image. This was performed using the CPLR based on the optimal SRII-RCSS. Thus, the co-registration between the previously generated normalized subject image and reference image was necessary to perform both methods.

The most important step in co-registration between an image pair is the selection of an appropriate mathematical model for transformation. Due to several factors, such as the large size of satellite images and the existence of complex image distortions, the global models (e.g., affine and projective) are usually insufficient for the accurate estimation of the transformation [43]. Consequently, in this study, a two-pass registration method based on the cost-effective B-spline function [45] was applied to obtain a high-precision registration of the normalized subject image to reference image. In the first pass, a simple 2-D linear model was used to estimate the affine transformation between the image pair at a coarse level using the generated corresponding points in step 1 (Section II-A1)

$$\begin{aligned} X &= a_0 x + a_1 y + a_2 \\ Y &= b_0 x + b_1 y + b_2 \end{aligned} \tag{11}$$

where $a_0 - a_2$ and $b_0 - b_2$ are the affine coefficients; and (X, Y) and (x, y) are the coordinates of match points in the reference image and in the normalized subject image, respectively. In the second pass, the registered normalized subject image was re-registered to the master image at a fine level using mapping parameters of the B-spline, which are obtained through the generated match points in step 2 (Section II-A2), as follows [45], [46]:

$$T(x, y) = \sum_{l=0}^{3}\sum_{m=0}^{3} B_l(u) B_m(\upsilon)\psi_{r+l,c+m} \tag{12}$$

where ψ denotes a $\delta_x \times \delta_y$ mesh of match points; r, c are the indices of the match point cell surrounding (x, y); (u, υ) denotes the relative position of (x, y) within the cell; $r = \lfloor (x/\delta_x) \rfloor - 1$, $c = \lfloor (y/\delta_y) \rfloor - 1$, $u = (x/\delta_x) - \lfloor (x/\delta_x) \rfloor$ $\upsilon = (y/\delta_y) - \lfloor (y/\delta_y) \rfloor$; and B_l is the basis function of B-spline which is given by the lth order of polynomials as follows [45], [46]:

$$\begin{aligned} B_0(u) &= (1-u)^3/6 \\ B_1(u) &= (3u^3 - 6u^2 + 4)/6 \\ B_2(u) &= (-3u^3 + 3u^2 + 3u + 1)/6 \\ B_3(u) &= u^3/6. \end{aligned} \tag{13}$$

The estimated $T(x, y)$ was used in this study to register the normalized subject image with the reference image. Additionally, the bilinear interpolation method was employed for resampling the normalized subject image to obtain a high registration accuracy.

To minimize the residual radiometric errors and possible differences, the normalization coefficients of the registered normalized image were updated through the CPLR method based on the optimal RCSS. To this end, let us consider the s_p^n is the DN values of the SRII-RCSS in the registered normalized images. By forming the critical region of CP process [(7)] based on the r_p and s_p^n, the updated coefficients were obtained as follows:

$$\alpha^u = \rho\frac{\sigma_r}{\sigma_{s^n}} \tag{14}$$

$$\beta^u = m_r - \alpha \times m_{s_n} \tag{15}$$

where, α^u and β^u are the updated slope and intercept, respectively; σ_{s_n} and m_{s_n} are respectively the standard deviation and mean of s_p^n; and the ρ is the correlation between r_p and s_p^n. The updated coefficients subsequently were used in the liner regression for generating the improved registered normalized image as follows:

$$S_i^{\mathrm{NI}} = \alpha_i^u S_i^N + \beta_i^u \tag{16}$$

where S_i^{NI} and S_i^N respectively refer to the DN values of the improved normalized image and original normalized image in ith spectral band.

B. Data Set Description

In this study, eight sets of bitemporal satellite images, acquired by different sensors at different geographic areas were used to comprehensively assess the efficiency of the

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TABLE I
CHARACTERISTIC OF SATELLITE DATA SETS USED IN THIS STUDY

Name	Ref./Sub. Image	Satellite/Sensors	Same spectral bands	Resolution (m)	Radiometric Resolution (bit)	Image Dimensions	Acquisition Date	Study Area
Dataset 1	Ref.	Sentinel-2 (MSI)	C/A, Blue, Green, Red, NIR, SWIR1, SWIR2	60/10/20	12	4663×4676	28 Jul 2019	Urmia Lake Iran,
	Sub	Landsat 8 (OLI)		30	16	4000×4000	20 Sep 2019	
Dataset 2	Ref.	SPOT 7	Blue, Green, Red, NIR, SWIR	2	16	4334×5907	01 Jan 2015	Venice, Italy
	Sub.	Pleiades		6		2975×3176	06 Jan 2017	
Dataset 3	Ref.	Landsat 5 (TM)	Green, Red, NIR	30	8	6981×7881	17 Feb 2007	Cape Town, South Africa
	Sub.	UK-DMC2		22		9300×9300	16 Feb 2012	
Dataset 4	Ref.	Landsat 7 (ETM+)	Green, Red, NIR, SWIR	30	8	2725×2949	21 Jun 2000	Ahvaz, Iran
	Sub.	Aster		15/30		5047×5671	3 May 2015	
Dataset 5	Ref.	Landsat 7 (ETM+)	Green, Red, NIR	30	8	4000×4000	13 May 2000	Barcelona, Spain
	Sub.	SPOT 5		10		5000×5000	21 May 2007	
Dataset 6	Ref.	SPOT 7	Blue, Green, Red	6	12	7000×7000	8 Aug 2014	Amman, Jordan
	Sub.	Bing Maps- satellite		16.34	8	3521×4129	19 Feb 2017	
Dataset 7	Ref.	IRS (LISS IV)	Green, Red, NIR	5.5	8	3000×3000	13 Jul 2007	Tabriz, Iran
	Sub.					3251×3251	5 Jul 2008	
Dataset 8	Ref.	Phantom 3 (UAV)	Blue, Green, Red	0.05	8	3624×2888	25 Dec 2017	Pardis-Tehran, Iran
	Sub.			0.03		4000×3000	24 Dec 2017	

C/A: Coastal/Aerosol, NIR: Near Infrared, SWIR: Short-wave Infrared

proposed RRN method (see Table I and Fig. 7). The data sets include six bitemporal intra-sensor (data sets 1–6) to investigate the performance of the proposed method in overcoming the typical RRN difficulties for multisensor acquisitions (e.g., different spatial resolutions and illumination geometries). The other two data sets (data sets 7 and 8) are bitemporal inter-sensor to demonstrate the potential of the proposed RRN method for the data sets acquired by the same sensor in different rows/paths. The selected data sets include no-data areas, cloud coverage, and large differences in spatial resolutions and illumination angles. It was also assumed that there was no information about the projection and coordinate systems for the subject image to evaluate the performance of the proposed method in situations where the non-georeferenced satellite images are available.

As is clear from Table I, the same spectral bands of the reference and subject image pair were used for RRN. The Landsat 5, 7, and 8, as well as Aster images, were downloaded from https://earthexplorer.usgs.gov/). The Sentinel 2 image was downloaded from the ESA's Sentinels data hub (https://scihub.copernicus.eu/dhus/#/home). The SPOT 5 and 7, U.K.-DMC2, and Pleiades imagery are openly available through the AIRBUS portal (https://www.intelligence-airbusds.com/en/5750-image-gallery-results). The scene of Bing satellite image was downloaded by SAS planet, a free open-source application. The spectral bands of data set 1 (Sentinel 2 image) with 20 and 60 m spatial resolutions were resampled to 10 m by the *Sen2Res* tool in the SNAP software. The SPOT 7 image used in Data set 6 was available in 8-bit radiometric resolution in the AIRBUS portal. The multispectral bands of data set 2 (SPOT 7 image) (R, G, B, NIR) with 6 m spatial resolution were pansharpened using the principal component analysis (PCA) technique [47] to improve the spatial resolution to 1.5 m. This method was also applied to the SWIR band of the ASTER image in data set 4 to generate imagery with 15 m spatial resolution.

C. Evaluation Criterion

In this section, various indices which were used for the quantitative analysis of the RRN results are provided. The mean and standard deviation values of the subject image, normalized subject image, and reference image, along with the normalized absolute error [NAE, (7)], structural content [SC, (18)], and peak signal to noise ratio [PSNR, (19)], were applied to compare similarity between the normalized subject image and its corresponding reference image for each spectral band using the test samples (i.e., 30% of the generated optimal SRII-RCSS which were same with IRMAD samples). It is worth noting that the test results for all RRN methods are computed over the same test samples to make fair comparisons

$$\mathrm{NAE} = \frac{\sum_{j=1}^{N_t} \left| R_j - S_j^N \right|}{\sum_{j=1}^{N_t} |R_j|} \tag{17}$$

$$\mathrm{SC} = \frac{\sum_{j=1}^{N_t} (R_j)^2}{\sum_{j=1}^{N_t} \left(S_j^N\right)^2} \tag{18}$$

$$\mathrm{PSNR} = 10 \log_{10} \left(\frac{(2^{n_b} - 1)^2}{\frac{1}{N_t} \sum_{j=1}^{N_t} \left(R_j - S_j^N\right)^2} \right) \tag{19}$$

where R and S^N indicate the reference and normalized subject images, respectively; N_t is the number of the test samples; and n_b is the number of bits/pixle. A lower value of NAE, closer value of SC to 1, and higher value of PSNR show a more accurate RRN result. The root mean square error [RMSE, (20)] was also employed to compare the efficacy of the proposed method with other state-of-the-art RNN techniques

$$\mathrm{RMSE} = \sqrt{\frac{1}{N_t} \sum_{j=1}^{N_t} \left(R_j - S_j^N\right)^2}. \tag{20}$$

Equation (21) which was calculated based on the checkpoint pairs (manually selected for each data set), was employed to evaluate the registration results

$$\mathrm{RMSE}_G = \sqrt{\frac{\sum_{i=1}^{N_c} \left[\left(X_i - X_i'\right)^2 + \left(Y_i - Y_i'\right)^2 \right]}{N_c - 1}} \tag{21}$$

where (X_i, Y_i) and (X_i', Y_i') are the coordinates of checkpoints in the reference and the transformed subject/normalized subject images, respectively; and N_c is the number of checkpoints.

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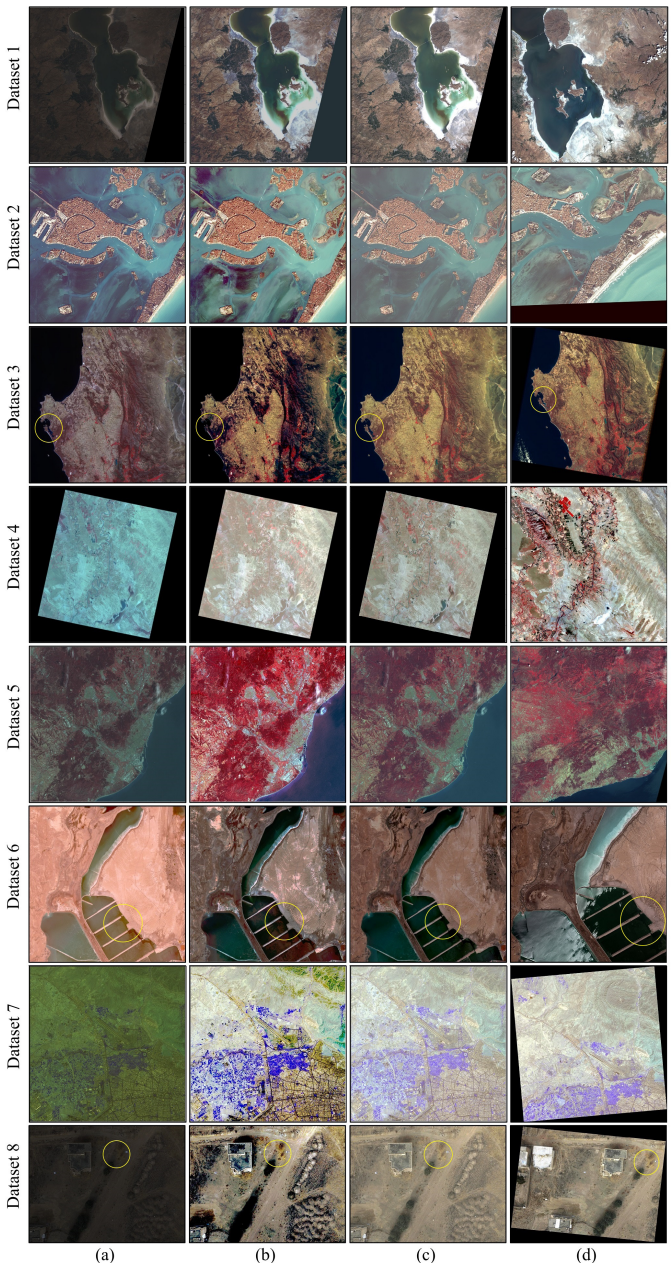

Fig. 7. Natural color composites (R/G/B) of data sets 1, 2, 6, and 8, false color composites (NIR/Red/Green) of data sets 3, 4, and 5, and false-color composites (Green/Red/NIR) of data sets 7. (a)–(d) Subject images, generated normalized subject images by HM, generated normalized subject images by the proposed method, and the reference images. Yellow circles indicate the regions which were selected to demonstrate the detailed information in the visual assessments.

III. Experimental Results and Analysis

The results of this study are provided in the following four subsections: 1) the procedure of selecting the RCSS using the proposed RRN method and the performance of CPLR are discussed; 2) the performances of the proposed method and state-of-the-art RNN techniques are assessed for the unregistered image pairs; 3) the performances of the proposed method and state-of-the-art RNN techniques are compared for registered image pairs; and 4) the effects of using down-sampling scales on the speed and accuracy of the proposed method

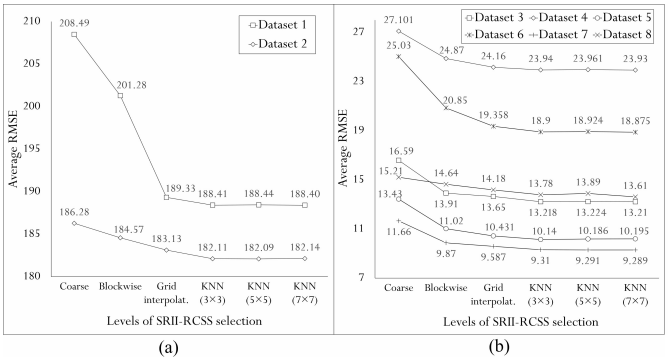

Fig. 8. Variation trends of average RMSEs obtained from each step of the SRII-RCSS selection by the proposed method for (a) data sets 1 to 2 and (b) data sets 3 to 8.

is investigated. The KAZE detector with the same set of parameters, recommended by Alcantarilla *et al.* [34], was used in this study for extraction of SRII-RCSS. Moreover, the block size was set to 250 × 250 pixels for the presented block-wise strategy to create a tradeoff between the number of corresponding points and the required runtime. The proposed method was implemented using Python 3.4, and the experimental results were obtained on a PC with Intel Core i7-3770 processor at 3.40 GHz, 16 Gigabytes of RAM.

A. Effects of Different Steps of the Proposed RRN Method

As discussed in Section II, the proposed RRN method uses a step-by-step procedure for selecting SRII-RCSS, including the coarse step (extracting global matched points in the coarse spatially alignment between image pair), the block-wise step (extracting match points in the block-wise KAZE matching strategy), the grid interpolation step (increasing the SRII-RCSS by regular grid interpolation), and KNN step (increasing the SRII-RCSS, extracted in the previous step, by considering nearest neighbor points). The variation trends of the average RMSEs on the spectral bands in selecting SRII-RCSS were calculated to evaluate the effectiveness of each step (see Fig. 8). It is worth noting that the CPLR was applied to obtain the normalized subject image in all steps, except step 1. This was because there was not enough SRII-RCSS for the CPLR and thereby ordinary linear regression (OLR) was used in the first step. Fig. 8 shows the downward trend of the average RMSE values after applying the step-by-step SRII-RCSS selection using the proposed RRN method. This trend started with a steep slope from the coarse step and ended with an almost steady trend at the KNN step. Generally, the proposed method had the best and worst efficiencies at the KNN and coarse steps for all data sets, respectively. In fact, the small number of SRII-RCSS extracted in the coarse step led to inaccurate estimation of the normalization coefficients and, thus, the normalization results were negatively affected. Then, as the number of SRII-RCSS in the block-wise step increased, the average RMSE decreased between 0.57 in the worst case and 7.20 at best for data set 5 and 8, respectively. As expected, increasing the number of SRII-RCSS using the grid interpolation method

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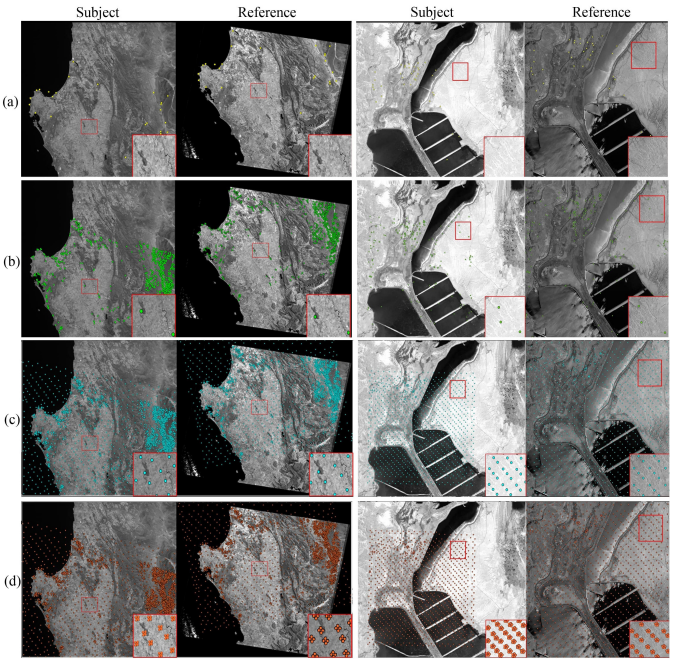

Fig. 9. Extracted SRII-RCSS by the (a) coarse, (b) block-wise, (c) grid interpolation, and (d) KNN steps for data set 3 (left column), and 6 (right column).

TABLE II

AVERAGE RMSES OF THE SPECTRAL BANDS OBTAINED FROM THE PROPOSED METHOD USING THE OLR, OR, RBR, RLR, AND CPLR TECHNIQUES

Dataset	Average RMSE						
	OLR	OR	RBR	RLR	CPLR		
					α=0.01	α=0.05	α=0.1
1	190.34	189.43	189.97	190.01	189.00	**188.41**	189.60
2	183.78	182.64	182.34	182.16	182.43	**182.10**	182.17
3	13.58	13.48	13.23	13.23	13.24	**13.22**	13.26
4	24.64	24.56	23.90	**23.87**	24.095	23.94	24.47
5	10.53	10.48	10.24	10.19	10.16	**10.14**	10.16
6	19.26	20.51	18.94	18.95	18.90	**18.89**	18.92
7	9.47	9.34	9.48	9.44	9.28	9.31	**9.24**
8	14.12	16.18	14.11	14.02	**13.76**	13.77	13.85

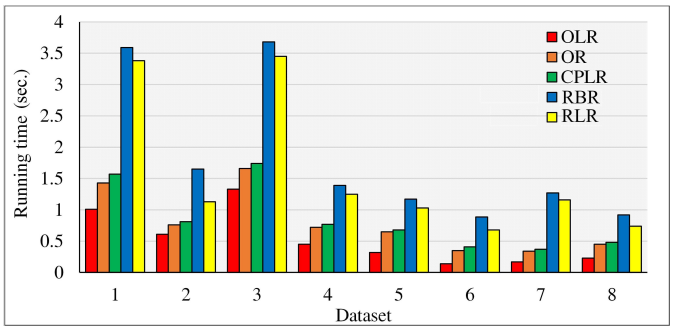

Fig. 10. Computation time of regression results on SRII-RCSS, generated by OLR, OR, CPLR with $\alpha = 0.05$, RBR, and RLR over all data sets.

reduced the normalization error in all data sets. The use of the grid interpolation step had the greatest effect on data sets 1, 2, and 6, where it respectively reduced the average RMSE values by 11.96, 1.44, and 1.49, compared to the block-wise step. This was mainly because the overlap regions within the image pair of these data sets mostly contained the homogeneous/textural-less areas (e.g. water areas), extracting SRII-RCSS over which was not accurately possible at the block-wise step due to the nature of feature detector methods (e.g., KAZE). However, in the grid interpolation step, in addition to the matches extracted in the block-wise stage, there was a uniform SRII-RCSS from textured and texture-less regions within the overlap regions of the image pairs. Therefore, this approach increased the flexibility of the proposed method and ensured the improvement of normalization accuracy in images with no or few textural information. Additionally, using the KNN approach resulted in the highest accuracy compared to other steps. For example, using 3×3 KNN respectively reduced the average RMSEs by approximately 0.97 and 0.34 for data sets 1–2 and data sets 3–8, compared to the grid interpolation step. However, increasing the neighborhood size from 3×3 to 5×5 or 7×7 did not always reduce the average RMSE. In this study, since increasing the neighborhood size reduced the computational efficiency, the 3×3 KNN was selected. Since the spatial distribution of RCSS has a significant effect on the RRN result, the extracted SRII-RCSS by the aforementioned steps were visually demonstrated in Fig. 9 along with the zoomed regions for data set 3 and 6.

As shown in Fig. 9, a few point pairs with the improper spatial distribution and positional shift errors were extracted in the coarse step, resulted in relatively higher error rates in the RRN procedure (i.e., higher RMSE in Fig. 8). The number of extracted matches from the overlap areas within the image pairs increased significantly in the block-wise step compared to the previous step. Furthermore, these match points were more properly distributed in the regions with high textural information. However, a few points were selected over the texture-less areas, such as deserts and water bodies, in this step [see Fig. 9(b)]. In fact, using the match points extracted in the block-wise step as SRII-RCSS could only yield acceptable normalization results for urban (e.g., data set 7) and mountainous (e.g., data sets 8) areas, which contained many textured regions. This was also confirmed where small differences between the quantitative results were observed in the normalization process using the block-wise step and the subsequent steps for data sets 7 and 8 (see Fig. 8). The grid interpolation step uniformly spread the SRII-RCSS over all LULC areas (textured and texture-less) within the overlap regions in the image pairs [see Fig. 9(c)]. Finally, using neighborhood information at each SRII-RCSS in the normalization process (i.e., a cluster of points are used instead of single points), had better distributions over the entire study areas [see Fig. 9(d)]. To evaluate the efficiency of the proposed CPLR method in different significance levels, its RRN results were compared with the RRN results of the OLR, Orthogonal regression (OR), and two robust regression based on the *bisquare* and *logestic* weighting functions, named as RBR and RLR in terms of average RMSEs and computational time (see Table II and Fig. 10). Moreover, the mentioned regression results on the SRII-RCSS extracted from the blue band of data set 8 were demonstrated in Fig. 11 for a line fitting assessment.

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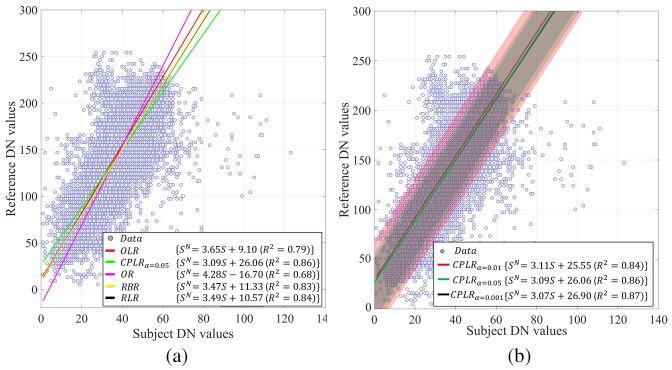

Fig. 11. Regression results, generated by (a) OLR, OR, CPLR with $\alpha = 0.05$, RBR, and RLR. (b) CPLR with $\alpha = 0.01$, 0.05, and 0.1 on the SRII-RCSS of the blue band of data set 8.

As is clear from Table II, the proposed CPLR method at different significance levels had the best performance as evidenced by smaller average RMSE in most cases, except for data set 4 in which RLR exhibited better average RMSE. For more detail, the OLR had the worst RRN performance with a high average RMSE for most data sets. This is mainly because the OLR method is highly sensitive to outliers, which often results in over/underestimation of normalization coefficients. Due to the consideration of measurement error both dependent and independent variables by the OR method [15], it provided more accurate RRN results than the OLR method overall. However, its estimates are deteriorated in some data sets (e.g., data set 6 and 8), indicating a sensitivity to outliers (see Table II and Fig. 11). On the other hand, the CPLR, RBR, and RLR had better RRN performance than the other ones in most cases by reducing the impact of outliers in estimating normalization coefficients. By comparing the CPLR method with RBR and RLR, it is observed that the proposed CPLR method was more stable and provided slightly better RRN results in a short time. This is mainly due to its adoption of the inexpensive CP process to estimate robust normalization coefficients (see Table II and Fig. 10). In general, there was not much difference between computational costs of OLR and OR or CPLR in modeling linear relationships between reference and subject images, whereas the RBR and RLR were much more computationally expensive (see Fig. 10). The results also showed that there was no significant difference in the performance of different significant levels of the CPLR method. However, the CPLR method with a significance level of $\alpha = 0.05$ had a relatively more stable performance in most cases, except for data sets 7 and 8.

A cursory look at Fig. 11 reveals that the CPLR method with the highest R^2 had the best performance in linear modeling of SRII-RCSS pairs in the presence of outliers, closely followed by RLR and RBR. By increasing the significance levels of CPLR method from $\alpha = 0.01$ to $\alpha = 0.1$, the R^2 values relatively increased [see Fig. 11(b)]. However, it should be noted that a high R^2 value does not necessarily mean the high accuracy of a model. For example, although considering a strict significance level, such as $\alpha = 0.1$, increased the R^2 values by reducing the critical region, it did not consider significant amounts of samples in determining the normalization coefficients and, thus, increasing the average RMSE values in most cases (see Table II). In contrast, when a smaller significance level was used (e.g., $\alpha = 0.01$), the risk of including outlier samples in the process of estimating normalization coefficients was raised by increasing the critical region [see Fig. 11(b)]. After considering all of these facts, CPLR with a significance level of $\alpha = 0.05$ was used in this study to have a balance in obtaining a high R^2 values and reducing the risk of including outliers in estimating normalization coefficients.

B. Comparative Results of the RRN Methods for Unregistered Image Pairs (Scenario 1)

Fig. 12 compares the PSNR, SC, NAE, mean, and standard deviation values of the reference and subject images with those of the reference and normalized subject images in the test samples using the proposed RNN method in this study. Fig. 12 shows the high capability of the proposed method in the RRN of the unregistered image pairs. The PSNR values for the normalized subject images were higher than those of the subject images in all spectral bands, indicating the better quality of the normalized images compared to the subject images in all data sets. Additionally, the measured SC values for the normalized images relative to the reference images were closer to 1 compared to those of the subject images relative to the reference images for all data sets. This indicated the high structural similarity of the normalized and reference images. Furthermore, the NAE values confirmed the similarity of the normalized and reference images across all spectral bands of data sets. Additionally, the mean values of the normalized and reference images were considerably close to each other in all bands of data sets, indicating the high similarity of the normalized and reference images [see Fig. 12 (d)]. In more detail, the proposed method had the least effects on the normalization of the blue, NIR, green, and red bands of data sets 6, 1/2, 3, and 4/5/7/8, respectively. In contrast, the proposed method had the greatest effect on the C/A, green, SWIR, red, blue, and NIR bands of data set 1, 2, 4/5, 6, 8, and 3/7, respectively.

As mentioned in Section I, spectral characteristics of LULC classes in the subject and reference images can be different due to several factors, such as variations in the atmospheric condition, soil color changes, and temporal and spatial differences in the geometry of illumination. Therefore, it was expected that the spectral characteristics of similar LULC in the normalized subject and reference images were close after applying the proposed RRN to the reference-subject image pair. To evaluate this, the spectral signatures for various LULC classes, including vegetation, bare soil, and water bodies were evaluated before and after the normalization, where the results are illustrated in Fig. 13. To this end, the mean values of pixels over multiple polygons from different LULC classes in the reference, subject, and normalized images were compared.

As can be seen from Fig. 13, the spectral signatures of LULC classes in the normalized images were closer to those obtained in the reference images for all data sets. However,

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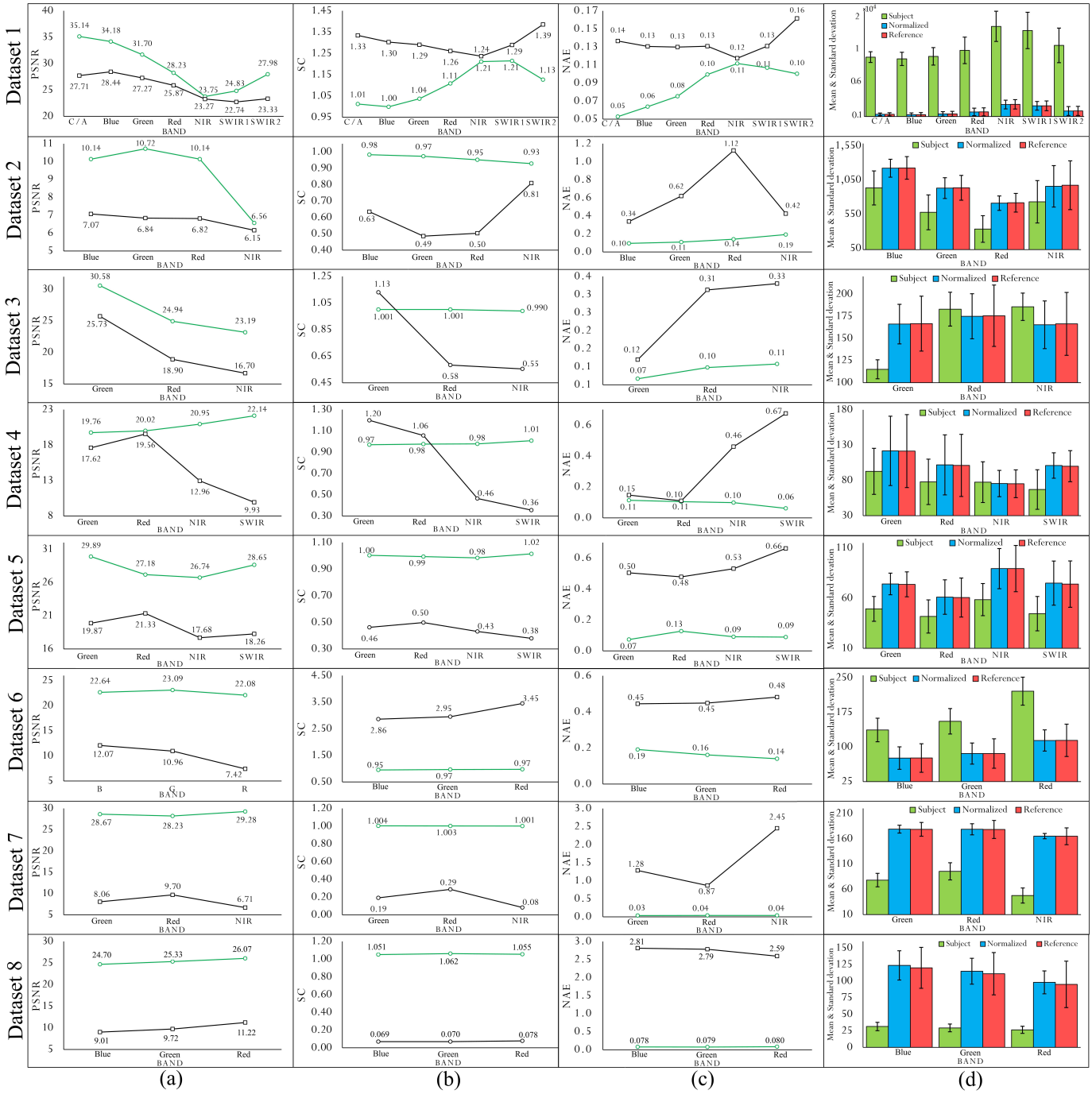

Fig. 12. (a)–(c) Comparison between the PSNR (dB), SC, and NAE values obtained over the test samples in the reference and subject images (black line), and between the reference and normalized subject images (green line). (d) Validation of the mean and standard deviation of the subject, normalized subject and reference images in all spectral bands for data sets 1 to 8.

the performance of the proposed method was not similar over all classes. For example, its performance in vegetation and bare soil areas was better than that of the water bodies. This was because the radiometric differences were only part of the factors affecting the spectral signature of water bodies, and sometimes other factors, such as water particles, marine pollution, currents, and phytoplankton changed the spectral characteristics (e.g., color) of water bodies. Moreover, a small shift was observed between the spectral signatures of vegetation covers in the reference and subject images due to differences in their phenological properties. For bare soil regions, the performance of the proposed RRN method was similar to that of the vegetation covers, and the differences in spectral signatures in the normalized and reference images were mainly due to characteristics of heterogeneous soils. Moreover, the difference in weather conditions and the acquisition times of the reference and subject images by different sensors were other factors of the difference in the spectral

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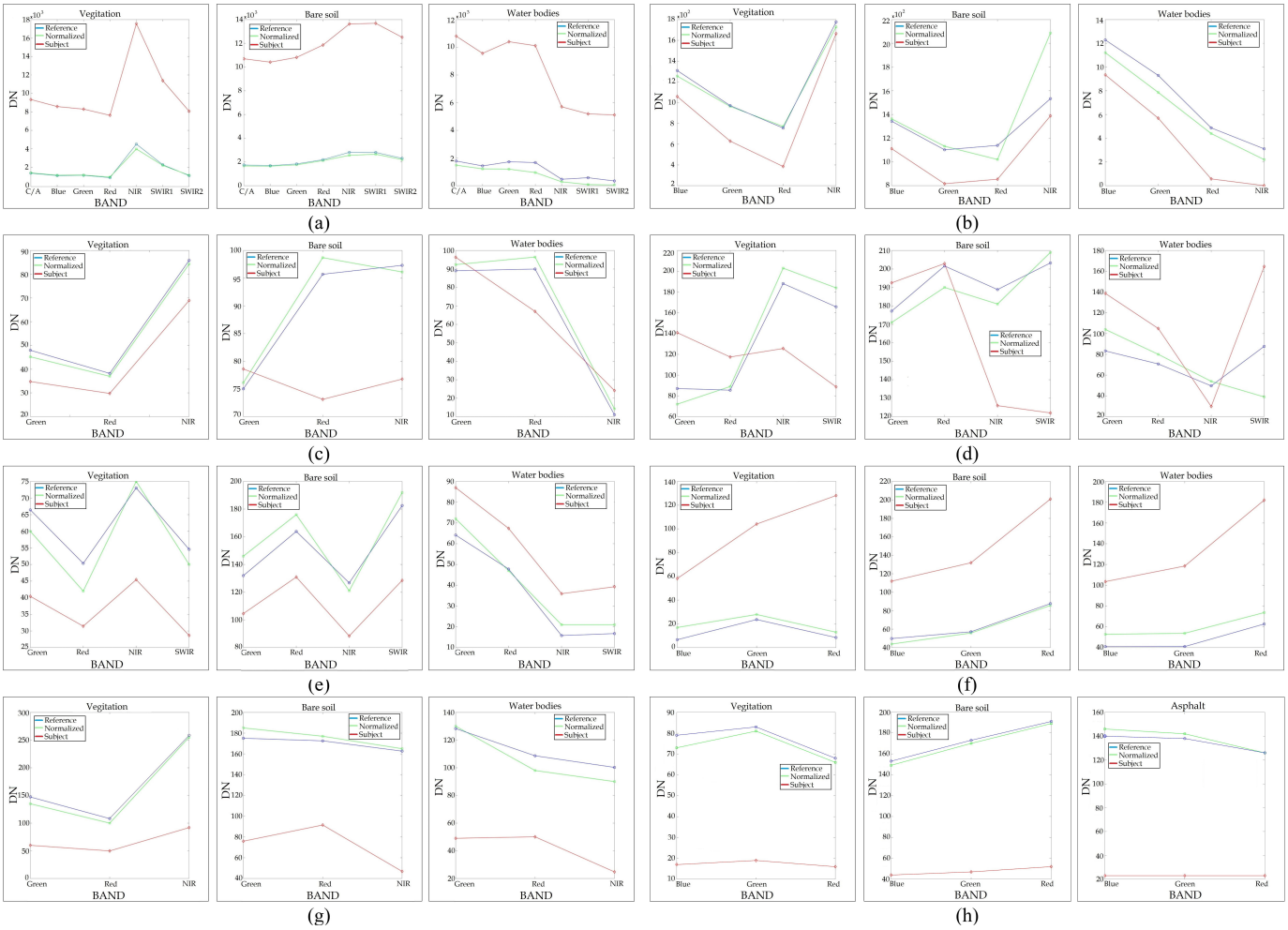

Fig. 13. Comparison between the spectral signatures of vegetation, bare soils, water bodies and asphalt (only for data set 8) before and after the normalization of data sets 1 (a)–8 (h).

signatures of similar LULC in the normalized and reference images. In summary, the characteristics of each LULC area in the normalized and reference images were similar, indicating the high efficiency of the proposed RRN method in reducing radiometric differences.

To assess the ability of the proposed method in RNN of unregistered image pairs, it was compared with the HM technique [17]. The quantitative and qualitative RNN results obtained from the proposed method and HM, as well as raw data for data sets 1–8 are illustrated in Table III and Fig. 7(a)–(d).

It can be seen from Table III that the proposed method provided superior RRN results in all spectral bands of data sets compared with the HM method. In fact, the proposed method was robust in terms of registration error due to utilizing the blockwise strategy to select SRII-RCSS in accurate positions in the reference and subject images. Moreover, Using CPLR within the proposed method allowed to create a precise relationship between DNs of SRII-RCSS in the subject and a reference image for all data sets. The proposed method improved the raw average RMSE values by 98.05%, 53.40%, 50.07%, 51.89%, 63.85%, 76.79%, 90.76%, and 83.05% for data sets 1 to 8, respectively. In contrast, the HM method not only increased the raw average RMSE for some of data sets (e.g., data sets 3 and 8) but also decreased the quality of the RRN results (see Fig. 7). For example, the HM increased the raw average RMSEs by 51.43% and 1.87% for data sets 3 and 8, respectively. Moreover, HM did not significantly improve the average RMSE of data set 2. This was mainly because the HM employs the global image statistics without considering LULC changes to match the histogram of the subject to that of the reference images [19].

As shown in Fig. 7, color, tone, and brightness of the normalized images generated by the proposed method were more similar to those of the reference images compared to the results of the HM algorithm for all of data sets. In fact, the HM algorithm created unwanted colors, noise, and artifacts in the normalized images, where these do not exist in the normalized image generated by the proposed method. Such distortions were mainly due to the nature of the HM method which tries to match the subject image histogram to the reference image histogram. Thus, the low similarity between the histogram of the subject and reference images led to a significant difference in the levels of grayscale between the reference image and the

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TABLE III

COMPARISON BETWEEN THE RRN RESULTS OF THE PROPOSED METHOD AND THE HM TECHNIQUE. THE RESULTS WERE ALSO PROVIDED WHEN RNN WAS NOT APPLIED TO DATA SET 1 TO 8 (I.E., RAW METHOD)

Dataset	Method	RMSE							
		C/A	Blue	Green	Red	NIR	SWIR1	SWIR2	Ava.
1	Raw	8722.67	8462.97	8591.04	9079.90	11993.62	11164.35	9615.50	9661.44
	HM	112.69	132.17	150.14	222.24	313.53	280.56	283.35	213.53
	Proposed Method	105.85	114.32	146.00	206.59	299.42	242.66	204.05	**188.41**
2	Raw	—	364.18	402.59	420.27	376.33	390.84	—	390.84
	HM	—	468.44	454.87	193.51	435.32	388.04	—	388.04
	Proposed Method	—	166.30	142.70	147.92	271.51	182.11	—	**182.11**
3	Raw	—	—	13.19	28.96	37.30	—	—	26.48
	HM	—	—	23.33	40.05	56.94	—	—	40.10
	Proposed Method	—	—	7.54	14.44	17.68	—	—	**13.22**
4	Raw	—	—	33.55	26.81	57.36	81.31	—	49.76
	HM	—	—	39.07	39.89	42.11	27.45	—	37.13
	Proposed Method	—	—	26.50	25.90	23.34	20.02	—	**23.94**
5	Raw	—	25.89	21.87	33.29	31.15	—	—	28.05
	HM	—	10.88	14.46	23.55	14.06	—	—	15.74
	Proposed Method	—	8.28	11.18	11.74	9.38	—	—	**10.14**
6	Raw	—	63.51	72.19	108.57	—	—	—	81.42
	HM	—	28.54	25.19	26.50	—	—	—	26.74
	Proposed Method	—	18.78	17.88	20.02	—	—	—	**18.90**
7	Raw	—	—	100.87	83.52	117.83	—	—	100.74
	HM	—	—	41.50	28.99	73.47	—	—	47.99
	Proposed Method	—	—	9.45	9.75	8.73	—	—	**9.31**
8	Raw	—	90.38	83.29	21.92	—	—	—	81.24
	HM	—	87.16	81.55	79.57	—	—	—	82.76
	Proposed Method	—	14.85	13.81	12.67	—	—	—	**13.78**

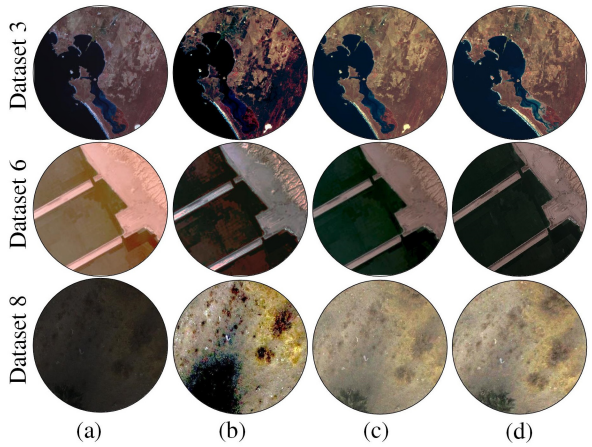

Fig. 14. Visual quality assessment of the proposed RNN and HM methods in the regions marked with the yellow circles in Fig. 7. (a) Subject images. (b) Normalized subject images by HM. (c) Normalized subject images by the proposed method. (d) Reference images.

normalized image generated by HM [see Fig. 7(b) and (d)]. For a more accurate visual comparison, Fig. 14 demonstrates the enlarged areas marked with a yellow circle from data sets 3, 6, and 8 in Fig. 7. As is clear from Fig. 14, there was more similarity between the vegetated (e.g., tree, grass, crop, and seaweed) and soil LULC in the reference image and generated normalized subject image by the proposed method. However, the contrast of these areas was dramatically increased in the normalized image generated by the HM algorithm and, consequently, the colors of these areas became darker. Such a contrast change in the normalized image generated by the HM method led to the loss of details in shadows and a potential false interpretation of land surfaces.

TABLE IV

COMPARISON BETWEEN THE REGISTRATION ACCURACY AND COMPUTATION TIME OF THE LOCAL AFFINE AND AFFINE-B-SPLINE, FOR DATA SET 1 TO 8

Data	Methods			
	Local affine		Affine-B-spline	
	RMSE (pixel)	Runtime (sec.)	RMSE (pixel)	Runtime (sec.)
Dataset 1	0.593	256.73	0.567	153.76
Dataset 2	1.013	168.56	1.003	97.31
Dataset 3	0.693	317.05	0.516	211.93
Dataset 4	0.825	177.66	0.715	91.43
Dataset 5	1.013	163.27	0.993	87.14
Dataset 6	1.320	256.13	1.194	146.21
Dataset 7	0.955	159.63	0.815	90.40
Dataset 8	1.624	141.47	1.594	52.16

C. Comparative Results of the RRN Methods for Co-Registered Image Pairs (Scenario 2)

In this section, the performance of the proposed RNN method is examined for registered image pairs. To this end, the affine-B-spline method was first used for each data set to register the normalized image to reference image using the extracted match points in steps 1 and 2 (Section II-A1 and II-A2). To evaluate the efficacy of the affine-B-spline method, its results were initially compared with those of the local affine techniques [43] using 40 evenly distributed checkpoint pairs which were randomly selected for each data set. The results of this assessment are provided in Table IV. It is worth noting that the calculated computational times include the sum of point extraction time as well as the transformation process.

It was observed from Table IV that the affine-B-spline method had overall the best performance, followed closely by the local affine method, indicating the capability of these two

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TABLE V

COMPARISON BETWEEN THE ACCURACY AND COMPUTATION TIME OF THE IRMAD, METHODS OF [33] AND [15], PROPOSED METHOD, AND IMPROVED PROPOSED METHOD. THE RESULTS WERE ALSO PROVIDED WHEN RNN WAS NOT APPLIED TO DATA SETS 1 TO 8 (I.E., RAW METHOD)

Dataset	Method	RMSE								Comp. time (sec.)	
		C/A	Blue	Green	Red	NIR	SWIR1	SWIR2	Ava.	Registration	RRN
1	Raw	8765.55	8516.87	8662.08	9172.03	12046.85	11273.44	9727.93	9737.82	153.76	—
	IRMAD	101.48	111.83	144.07	193.74	304.96	221.99	185.17	180.46		166.65
	Method of [33]	103.16	107.62	155.84	201.82	298.14	237.50	194.41	185.50		149.36
	Method of [15]	102.45	116.94	163.28	226.63	286.48	223.22	182.63	185.95		205.17
	Proposed method	92.75	104.44	134.22	187.20	285.18	218.90	180.09	171.83		**23.24**
	Improved Proposed method	92.25	103.90	133.34	186.12	285.02	218.27	179.25	**171.16**		28.41
2	Raw	—	367.89	406.67	425.49	386.13	—	—	396.55	97.31	—
	IRMAD	—	174.59	150.02	164.10	344.58	—	—	208.32		152.31
	Method of [33]	—	171.89	154.92	170.27	288.48	—	—	196.39		130.17
	Method of [15]	—	173.38	151.44	162.82	288.79	—	—	194.11		194.13
	Proposed method	—	173.33	151.32	160.56	286.32	—	—	192.88		**22.03**
	Improved Proposed method	—	172.29	150.98	159.71	284.74	—	—	**191.93**		26.95
3	Raw	—	—	13.12	28.83	37.27	—	—	26.41	211.93	—
	IRMAD	—	—	7.68	15.30	18.24	—	—	13.74		152.31
	Method of [33]	—	—	7.67	14.94	18.13	—	—	13.58		173.07
	Method of [15]	—	—	7.61	14.81	18.06	—	—	13.49		280.76
	Proposed method	—	—	7.48	14.21	17.61	—	—	13.10		**22.03**
	Improved Proposed method	—	—	7.46	14.16	17.60	—	—	**13.07**		26.95
4	Raw	—	—	33.35	26.50	57.39	81.40	—	49.66	91.43	—
	IRMAD	—	—	28.82	28.20	25.73	20.67	—	25.85		46.39
	Method of [33]	—	—	26.47	26.31	27.54	21.81	—	25.53		37.26
	Method of [15]	—	—	26.33	25.50	23.13	19.78	—	23.68		79.35
	Proposed method	—	—	25.64	24.90	22.40	19.84	—	23.19		**15.79**
	Improved Proposed method	—	—	25.56	24.88	22.40	19.60	—	**23.11**		18.27
5	Raw	—	—	26.28	23.47	34.47	32.41	—	29.16	87.14	—
	IRMAD	—	—	10.05	15.90	16.51	20.90	—	15.84		59.68
	Method of [33]	—	—	11.55	16.69	15.57	15.14	—	14.74		51.23
	Method of [15]	—	—	10.85	16.47	15.18	15.63	—	14.53		106.04
	Proposed method	—	—	11.34	15.99	14.88	15.13	—	14.34		**15.31**
	Improved Proposed method	—	—	10.24	15.01	14.71	14.44	—	**13.60**		17.54
6	Raw	—	64.19	72.94	109.34	—	—	—	82.16	146.21	—
	IRMAD	—	23.39	25.03	29.98	—	—	—	26.13		157.26
	Method of [33]	—	23.51	23.05	26.11	—	—	—	24.22		153.16
	Method of [15]	—	24.72	24.40	27.46	—	—	—	25.52		220.15
	Improved Proposed	—	21.69	21.53	25.36	—	—	—	22.86		**22.13**
	Improved Proposed method	—	21.27	21.00	24.34	—	—	—	**22.21**		26.69
7	Raw	—	—	100.93	83.58	117.86	—	—	100.79	90.4	
	IRMAD	—	—	12.29	10.59	8.58	—	—	10.49		48.12
	Method of [33]	—	—	9.11	8.96	8.05	—	—	8.71		42.19
	Method of [15]	—	—	8.13	8.24	8.01	—	—	**8.13**		89.13
	Proposed method	—	—	8.23	8.63	7.99	—	—	8.28		**15.76**
	Improved Proposed method	—	—	8.16	8.63	7.99	—	—	8.26		17.78
8	Raw	—	86.55	79.35	66.02	—	—	—	77.31	52.16	—
	IRMAD	—	19.13	18.24	16.74	—	—	—	18.04		41.15
	Method of [33]	—	18.90	16.58	15.38	—	—	—	16.95		38.18
	Method of [15]	—	18.25	17.06	15.72	—	—	—	17.01		67.65
	Proposed method	—	17.80	16.56	15.56	—	—	—	16.96		**11.19**
	Improved Proposed method	—	16.61	15.55	14.70	—	—	—	**15.62**		16.26

approaches to model the existing complex distortions between image pairs acquired by the inter/intra sensors. However, since the local affine technique had more computational cost, the affine-B-spline method was used in this study. After registration of the normalized subject image to the reference image, its normalization coefficients were updated for each band based on the CPLR to generate the improved normalized image. Hereinafter, the method proposed to generate improved normalized images using CPLR was named the improved proposed RNN method. To evaluate the capability of the proposed RNN method and its improved version against SRRN methods, their results were compared with those of the IRMAD [22], and the most recent methods proposed by Ghanbari *et al.* [33] and Moghimi *et al.* [15] using the same set of parameters as recommended in [15], [22], and [33]. For this, the subject and normalized subject images were registered to the corresponding reference images using the affine-B-spline method. Then, the registered images were applied to generate RRN results using the IRMAD, methods of [33] and [15], proposed RNN method, and improved proposed RNN method. The quantitative results obtained from this assessment are provided in Table V. Additionally, the chess-board visualization of the data sets, before and after normalization by the aforementioned RRN methods, was generated for qualitative assessment, where the results are illustrated in Fig. 15.

As can be seen from Table V, the improved proposed method provided superior RRN results in most cases, except for data set 7 in which method of [15] yielded lower average RMSE. For instance, the improved proposed RNN method reduced the raw average RMSEs by 98.24%, 51.60%, 50.51%, 53.46%, 53.36%, 72.96%, 91.80%, and 79.79% for data

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sets 1–8, respectively. Furthermore, the remaining radiometric differences as well as DN changes due to wrapping operation (e.g., bilinear) from the normalized image generated by the proposed method slightly reduced after using its improved version (i.e., approximately less than 5%). The IRMAD and methods Ghanbari *et al.* [33] and Moghimi *et al.* [15] provided very similar RRN results with less than 1% performance degradation on the average with respect to the proposed approach and its improved version in RRN of data sets 4 and 7. Meanwhile, the RRN results from the methods of [33] and [15] showed relatively 14% performance degradation in the worst case (on data set 1) with respect to the proposed method and its developed version. In addition, the performance degradation in the IRMAD method compared to the improved proposed method was more obvious for data sets 1 and 2 with the average RMSE differences of 9.3 and 16.39, respectively. The methods [33] and [15] were more accurate than IRMAD in RRN of most cases because they include outlier refinement step in their RRN procedure. Generally, the IRMAD method and methods [33] and [15] had fluctuating performance in RRN of the multiple data sets, especially for multisensor data sets (i.e., data sets 1–6). However, both the proposed method and its improved version by CPLR had a reliable performance. This was mainly because the performance of IRMAD and methods [33] and [15] negatively affected by differences in illumination and spatial resolutions between image pairs. However, the proposed RNN method was independent of these distortions due to the use of the potential of the KAZE detector for the SRII-RCSS selection.

Considering similar registration time for all the methods, it was observed that the proposed method and its improved version were much faster than the other methods with the least computational cost. The method of [15] had the highest processing time for RRN of all data sets (see Table III), due to the use of multiple step for the RCSS selection. Moreover, the method of [33] was computationally efficient compared with the IRMAD and method of [15]. In the worst and best cases, the improved proposed method was 21.92 and 146.12 s faster than the method of [33] for data set 8 (with a size of $3624 \times 2888 \times 3$) and data set 3 (with a size of $6981 \times 7881 \times 7$), respectively. The reason for the low time complexity of the proposed method before and after improvement was that the match feature points generated in the registration step were employed as SRII-RCSS. In contrast, the whole image space was inspected to find RCSS by the IRMAD and methods of [33] and [15], resulting in the high computational complexity. Finally, it should be noted that the runtime of the proposed method increased by 13% to 32% after improvement (i.e., updating the normalization coefficients). Therefore, if a wide range of radiometric differences could be eliminated or reduced by the proposed method, there would be no need to update the normalization coefficients using CPLR.

The results illustrated in Fig. 15 showed that the brightness intensity errors between slices were significantly reduced after the normalization by the proposed method, and its improved version, where the smooth continuation was observed between the reference and normalized subject slices. In other words, more seamless image chessboards were observed after normalization by the mentioned RRN methods, and, therefore, the vignette effect (local color inconsistency) was minimized. There was not much visual difference between the normalized images generated by the proposed method and its improved version. This was because most of the radiometric differences between the subject and reference images were already removed by the proposed method. In more detail, the IRMAD method, methods of [33] and [15], and the proposed method had very similar performance in term of visual inspection for data sets 7 and 8 which were acquired from the same sensors [e.g., see last two columns of Fig. 15(a)–(f)]. Moreover, the methods of [33] and [15] were much visually accurate than IRMAD and had the almost same performance as the proposed method for data set 2 and 3 [e.g., see columns 2 and 3 in Fig. 15(a)–(f)]. However, the performance of the proposed method before and after improvement was overall better than them in most of data sets which were acquired from different sensors [e.g., see columns 4, 5, and 6 in Fig. 15(a)–(f)]. This indicated the efficacy of both scenarios of the proposed method in RRN of the multisensor image pairs. For example, vegetation areas in the normalized image of data set 1, which was generated by IRMAD and methods of [33] and [15] were darker than those in the reference image [see first column of Fig. 15(a)–(f)], where there were good agreements between slices of reference and subject images. As another example, the brightness of the normalized image generated by the IRMAD method was abnormally increased in data set 2 and, thus, a huge difference was observed in the slices of the corresponding chessboard (see the second row of Fig. 15). These were mainly related to the incorrect estimation of the normalization coefficients by the IRMAD method and, thus, outliers were not accurately considered in the modeling of the relationship between the reference and subject images. This limitation increased the difference in intensity values between the slices in the chessboards generated by the IRMAD method. Visually, the proposed method and its improved version had a moderate performance on data set 6 compared to other data sets, where the brightness differences between slices were more obvious in the chessboards. This was due to the sharp contrast differences between the reference and subject images and their different acquisition times [see column 6 of Fig. 15(a)–(f)].

D. Effects of Down-Sampling Scales

As mentioned in Section II, down-sampled image pairs were used in the experiments (i.e., matching process) to reduce computational complexity and to save time. Then, the coordinates of the extracted match points were transferred to the image with the original size to be used in the process of normalization and estimation of the final 2-D transformation. In this study, the effects of down-sampling at the scales of 0.25, 0.5, 0.75, and 1 on data set 1 were obtained (see Table VI) to investigate the effects of the down-sampling scales on the amount of computation time, as well as on the quality of the RRN. The CPU consumption of the system based on different down-sampling scales is also demonstrated in Fig. 16.

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Fig. 15. Chessboard visualizations of data sets 1–8 to evaluate the RRN results of the reference-subject image pairs (a) before the normalization, and after the normalization by (b) IRMAD method, (c) method of Ghanbari *et al.* [33], (d) method of Moghimi *et al.* [15], (e) proposed method, and (f) improved proposed method. The white slices of the chessboard indicate the subject or normalized subject images and the black slices show the reference images.

TABLE VI

NUMBER OF MATCH POINTS, AVERAGE RMSE, AND TOTAL COMPUTATION TIME WHICH RESULTED FROM THE PROPOSED METHOD WITH DIFFERENT DOWN-SAMPLING SCALES APPLIED TO DATA SET 1 (WITHOUT THE REGISTRATION PROCEDURE)

Scale	Number of matches	Average RMSE	Computation time (Sec.)
0.25	677	193.1125	130.04
0.5	2312	188.4181	174.13
0.75	2850	188.1305	319.34
1	4759	188.0905	812.17

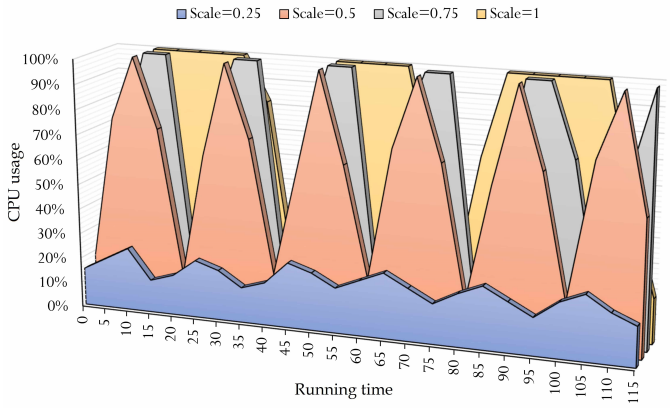

Fig. 16. CPU usage comparison resulted from the proposed method with different down-sampling scales applied to data set 1.

Based on the results provided in Table VI, although the numbers of matches were almost quadrupled by changing the scale value from 0.25 to 0.5, the time of the point extraction was increased by 40 seconds. However, increasing this number of matches on the 0.5 scale reduced the average RMSE of RNN by 4.69. In fact, many details of the image were lost at the scale of 0.25, which reduced the number of extracted features from each image. Moreover, although the numbers of matches were almost doubled by changing the scale from 0.5 to 1, the required time for feature extraction and matching increased more than five times and the rate of change in average RMSE was insignificant (0.32). In fact, the scales of 1 and 0.75 not only did not significantly increase the

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RRN accuracy but also increased the time and computational complexity of the RRN process, which was in line with Fig. 16.

IV. Conclusion

The RRN of the multitemporal and multisensor remote sensing images is a fundamental preprocessing step for many remote sensing applications. Traditional SRRN methods do not have satisfactory results for unregistered multisensor image pairs with different rotations, illumination geometries, and spatial resolutions. Therefore, this study introduced a novel method for RRN of multitemporal and multisensor remote sensing images mainly based on utilizing the KAZE detector and CPLR method. The SRII–RCSS were first selected by a block-wise strategy using the KAZE detector. They were evenly distributed in the textured and texture-less LULC areas using the grid interpolation and KNN methods and then were refined based on a new similarity measure to extract appropriate SRII-RCSS candidates. The normalized image was generated by applying the CPLR to reference–subject image pair using the optimal SRII-RCSS candidates. Finally, the normalized image was registered to the reference image through the affine-B-spline method and then was enhanced by updating its normalization coefficients based on the CPLR method.

Several inter/intra sensors image pairs were utilized to evaluate the performance of the proposed method in RRN of unregistered images. Experimental results showed that using the corresponding points generated from the block-wise KAZE strategy had great potential in addressing difficulties in RRN of the unregistered multisensor image pairs. Dense and well spatially distributed SRII-RCSS were generated in both texture-less and textured LULC areas using the grid interpolation and KNN methods. The possible changed pixel pairs within the SRII-RCSS were accurately refined based on the proposed score index. The CPLR provided better results than OLR, OR, RBR, and RLR for estimating the linear relationship between the reference and subject images due to utilizing the CP process for removing outliers during the linear fitting. Comparing the PSNR, SC, and NAE values of the reference and subject images to those of the reference and normalized subject images in the test samples verified that the differences between reference and subject images decreased significantly after the normalization using the proposed method. The mean and standard deviation values of the normalized and reference images were also close to each other, demonstrating a high similarity between these images. The results showed that the proposed RNN method outperformed the HM technique in terms of quantitative and qualitative accuracies. This was due to using the SRII-RCSS during the proposed RRN procedure. In RRN of registered image pairs, the proposed method and its improved version provided more stable results than the IRMAD technique and methods of [33] and [15], and they effectively reduced the radiometric differences from the subject images. To simplify the modeling process in this study, it was theoretically assumed that the relationship between the spectral response of SRII-RCSS at two different times can be estimated by a linear function. However, this hypothesis is unsuitable in dealing with image pairs with significant LCLU changes, whereas nonlinear models are more robust [18]. Therefore, the potential of different non-linear mapping functions in modeling the relationship between the image pair based on the SRII-RCSS will be studied in near future.

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